

# Adaptation conditions

**Preset:** celltype\_pairs\_Sc0p2\_noise0p025\_dualStd\_7cond · **Model class:** SRNNCellTypePairs

GENERATED by `write_manuscript_tables` from a model BUILT from that preset under the **sfa\_and\_std** condition. Do not edit by hand – every value here is read off the object, so it cannot disagree with what the sweeps ran. Regenerate after changing the preset.

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Every sweep runs each grid point under all 7 regimes, on the **same network** (the weight seed is shared), so the comparison is paired.

| Condition         | SFA timescales $\tau_a$ (s) | Depressing routes                              | Facilitating routes |
|-------------------|-----------------------------|------------------------------------------------|---------------------|
| No Adaptation     | none                        | none                                           | none                |
| SFA (1 tau)       | 0.25                        | none                                           | none                |
| SFA Only          | 0.25, 1.581, 10             | none                                           | none                |
| STD (1 tau)       | none                        | E to E, E to I, I to E, I to I (tau_rec 2)     | none                |
| STD Only          | none                        | E to E, E to I, I to E, I to I (tau_rec [2 4]) | none                |
| SFA + STD (1 tau) | 0.25, 1.581, 10             | E to E, E to I, I to E, I to I (tau_rec 2)     | none                |
| SFA + STD         | 0.25, 1.581, 10             | E to E, E to I, I to E, I to I (tau_rec [2 4]) | none                |

**Why this class.** SRNNCellTypePairs names each synaptic route individually, so depression can be put on one route and not another. SRNNModel2 cannot express that: its depression count is per **presynaptic population**, so it can say “all outgoing E synapses depress” but never distinguish E to E from E to I.

**Implementation note.** When a mechanism is switched off its state variables are excluded from the state vector and from the Jacobian entirely, rather than being integrated at zero. This prevents spurious zero eigenvalues from disabled dynamics.